

SPEECH UNDERSTANDING

(CSL7770)

Paper Review

ASSIGNMENT 3

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MFF-EINV2: Multi-scale Feature Fusion across Spectral-Spatial-Temporal Domains for Sound Event Localization and Detection

(Da Mu, Zhicheng Zhang, Haobo Yue)

SUMMARY

This paper introduces a novel model called MFF-EINV2 [1] for **Sound Event Localization and Detection** (SELD), a task that involves identifying what sound events occur in an audio scene, when they occur, and from which direction they come. The proposed model builds upon an existing method known as EINV2, which has already demonstrated solid performance in the field. However, EINV2 and similar models often rely on simple CNN or ResNet stacks, limiting their ability to fully capture features across the spectral (frequency), spatial (location), and temporal (time) dimensions. The authors aim to improve feature extraction across these three critical domains:

1. **Spectral domain**: related to sound frequencies (e.g. high vs. low tones),
2. **Spatial domain**: where in space the sound is coming from,
3. **Temporal domain**: how sound evolves over time.

To address these limitations, the authors design a new Multi-scale Feature Fusion (MFF) module, which enables the network to extract features at multiple scales from each domain using parallel subnetworks and a TF-Convolution Module (TFCM).

Like EINV2, MFF-EINV2 adopts a dual-branch architecture:

1. The **Sound Event Detection branch**: identifies what sound is occurring.
2. The **Direction of Arrival branch**: estimates where the sound is coming from.

These branches are linked through soft parameter sharing, meaning they share internal representations to better learn both tasks jointly.

This paper introduces three key modules that significantly enhance feature extraction across spectral, spatial, and temporal domains:

1. Parallel Multi-resolution Subnetworks

The MFF module uses three subnetworks operating at different frequency resolutions to capture both fine-grained spectral details and broader patterns. This setup also improves spatial representation for better sound localization.

2. TF-Convolution Module (TFCM)

Each subnetwork includes TFCMs that apply dilated convolutions to extract multi-scale temporal features (from short, transient sounds to longer, continuous ones).

3. Repeated Multi-scale Fusion

Features from different subnetworks are fused repeatedly across stages, allowing cross-scale learning and richer representations. Frequency down sampling and up sampling ensure feature maps align for effective fusion.

These innovations help the model better detect overlapping sounds and track their evolution over time and space.

The final model, MFF-EINV2, not only **improves performance** by 18.2% on the SELD score metric but also **reduces the number of parameters** by 68.5% compared to EINV2, making it more efficient. It **achieves state of the art results** on benchmark datasets (STARSS22 and STARSS23 from the DCASE challenge), with notable improvements in both detection accuracy and localization precision (all achieved without any data augmentation).

ARCHITECTURE OF MFF-EINV2

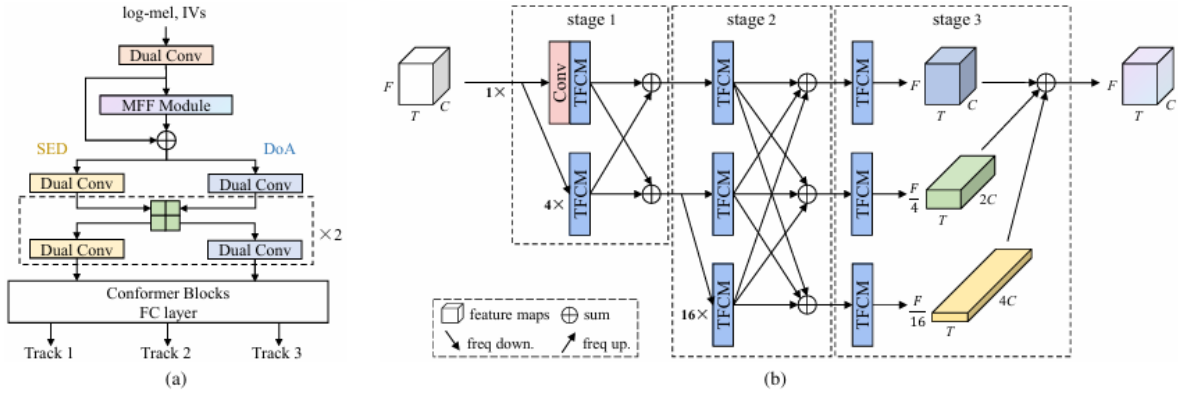


Figure 1 Illustration of the overall MFF-EINV2 architecture and the internal structure of the MFF module.

a) MFF-EINV2 Pipeline Diagram

This diagram shows the full model pipeline. The input consists of log-Mel spectrograms and intensity vectors (IVs), which are processed by a Dual Convolution (Dual Conv) layer, followed by the proposed MFF Module. The output of the MFF Module is fed into two parallel branches:

- i. The SED branch (in yellow) for Sound Event Detection.
- ii. The DoA branch (in blue) for Direction of Arrival estimation.

Both branches consist of additional Dual Conv layers. The green box indicates soft parameter sharing between branches, allowing shared learning across both tasks. Outputs from both branches are passed into a shared sequence of Conformer Blocks and a final Fully Connected (FC) layer, which produces three track-wise outputs for SELD tasks.

b) MFF Module Details

This module is structured in three stages. Each stage contains three parallel subnetworks ($1\times$, $4\times$, $16\times$), operating at different frequency resolutions:

- i. $1\times$: Original resolution (full frequency range).
- ii. $4\times$ and $16\times$: Lower resolutions created through frequency down sampling (FD) to capture broader spectral patterns.

Each subnetwork uses a TF-Convolution Module (TFCM) to extract temporal and spectral features at multiple scales. Arrows across subnetworks represent cross-resolution connections, facilitating repeated multi-scale feature fusion. After the third stage, features from different scales are up sampled (FU) and concatenated into a unified feature

representation, enhancing the model’s ability to combine fine and coarse features across time, frequency, and space.

STRENGTHS OF THE PAPER

1. Multi-scale Feature Extraction

The MFF module extracts features at various temporal, spectral, and spatial scales, capturing both local details and global patterns. Many SOTA methods use fixed-scale feature extraction, which can miss important variations. Multi-scale processing enables better detection of overlapping or complex sound events.

2. Parallel Multi-resolution Subnetworks

Three subnetworks working at different frequency resolutions ($1\times$, $4\times$, $16\times$) allow the model to understand fine spectral textures and broad trends simultaneously. This helps outperform traditional CNN-based encoders that operate at a single resolution, making the model more versatile in handling diverse audio scenes.

3. TF-Convolution Modules (TFCMs)

TFCMs use dilated convolutions to learn both short-term and long-term temporal dependencies within each frequency band. SOTA models often struggle to balance rapid and slow-evolving sound events. TFCMs address this by flexibly modelling both, improving event detection and duration estimation.

4. Repeated Multi-scale Fusion

Features are fused across subnetworks after each stage, encouraging information sharing and refining representations. This enables the network to combine strengths from different scales repeatedly, improving the consistency of detection and localization (a limitation in previous static-fusion methods).

5. Soft Parameter Sharing Between Branches (SED & DoA)

The SED and DoA branches share parts of the network, allowing them to learn from each other while optimizing for their specific tasks. This improves multi-task learning efficiency and ensures tighter coupling between what sound is happening and where it’s coming from (essential for accurate SELD).

6. Efficient Architecture with Reduced Parameters

Despite performance gains, the model uses 68.5% fewer parameters than EINV2. This efficiency is critical for deploying SOTA models in real-time or resource-limited environments (e.g., smart devices or edge computing).

7. SOTA Results Without Data Augmentation

The model achieves state of the art scores on STARSS22 and STARSS23 datasets without needing any synthetic data or augmentation. This demonstrates strong generalization and robustness, unlike many SOTA systems that heavily rely on augmentation to achieve competitive performance.

WEAKNESSES OF THE PAPER

1. Lack of Analysis on Real-World Noise Robustness

The paper does not thoroughly evaluate how the model performs under varying levels or types of real-world background noise. SELD applications often deal with noisy environments (e.g., urban or home settings). A SOTA method should test generalization under such real-world variability.

2. Limited Ablation on Fusion Strategies

Although the methodology described in the paper uses repeated multi-scale fusion, it does not extensively compare this with alternative fusion methods (e.g., attention-based fusion or late-stage aggregation). Without detailed ablation, it's unclear if repeated fusion is optimal or if other methods could achieve similar performance with lower complexity.

3. Increased Architectural Complexity

While the model reduces computational overheads by a good fraction by reducing the number of trainable parameters, the inclusion of parallel subnetworks, repeated fusion stages, and TFCMs increases the model's architectural complexity which may make it harder to adapt, debug, or deploy in highly constrained real-time systems.

4. Soft Parameter Sharing Details Underspecified

It's unclear how much the soft parameter sharing between SED and DoA branches contributes to performance, and whether hard sharing or independent branches would perform better or worse. An analysis of the relation between the two branches could have helped more in understanding the impact of soft parameter sharing on joint learning performance, and whether stronger coupling or full independence would lead to better optimization for SED and DoA tasks.

MINOR QUESTIONS / MINOR WEAKNESSES

1. Limited Generalization Evidence Across Datasets

Results are only shown on STARSS22 and STARSS23, the paper doesn't test the model on other SELD datasets (e.g., TAU, Ambisonic datasets). Evaluating across diverse datasets could help demonstrate broader.

2. Effect of Input Features Not Explored

The model uses log-Mel spectrograms and IVs, but the effect of using other features (e.g., raw waveforms or phase information) isn't tested. Using different input modalities could further boost or simplify performance.

3. No Comparison with Other Temporal Modelling Techniques

While the TF-Convolution Module (TFCM) is effective, the paper does not compare it with alternative temporal modelling methods like Temporal Convolutional Networks (TCNs) or Transformer-based temporal blocks. A brief comparison could validate the choice.

4. Lack of Analysis on Convergence Speed or Training Stability

The paper does not report how quickly or stably the model converges during training. Including training curves or convergence behaviour could help practitioners assess model efficiency.

5. Fusion Strategy Focused Only on Frequency Domain

The fusion primarily aligns feature maps along the frequency dimension. Exploring whether temporal or channel-wise alignment could improve performance would be a worthwhile extension.

REVIEW

1. SUGGESTION

To strengthen the paper, I recommend the authors evaluate the model's robustness under varying real-world noise conditions, as SELD systems are frequently deployed in unpredictable acoustic environments. Incorporating ablation studies that compare the proposed repeated multi-scale fusion with alternative strategies, such as attention-based or late-stage fusion would provide stronger support for the chosen design. Furthermore, offering empirical insights into the role and impact of soft parameter sharing between the SED and DoA branches would improve understanding of the model's joint learning dynamics. Analysing convergence behaviour and reporting training stability could also be helpful for future reproducibility and deployment.

2. RATING: (8/10)

The paper proposes a technically solid and efficient SELD model with state-of-the-art performance and clear innovation. A few additional evaluations would further solidify its claims, but the contribution is significant and well-executed.

BONUS QUESTION

(Reproduce the results of the paper on any 2 datasets mentioned in the paper.)

Reproducing the full results of the paper on two datasets was computationally very expensive. Due to limited time and computational resources, I modified some of the hyperparameters mentioned in the paper to reduce the training load. Specifically, I reduced the number of epochs from 100 to 10 and the batch size from 6 to 4. While these changes made the experiments more feasible within the available resources, they also led to a slight degradation in performance compared to the results reported in the paper.

The two datasets used for this purpose were STARSS22[2] (Sony-TAU Realistic Spatial Soundscapes 2022 dataset) and STARSS23[3] (Sony-TAU Realistic Spatial Soundscapes 2023 dataset), both of which are part of the DCASE Challenge. These datasets consist of realistic spatial soundscapes recorded in various indoor environments using First-Order Ambisonics (FOA) microphone arrays. They include polyphonic sound events with precise temporal and spatial annotations, making them highly suitable for evaluating Sound Event Localization and Detection (SELD) systems.

1. Results on STARSS22

SELD Class-wise Evaluation Results

Class ID	Error Rate (ER)	F-score (F)	Localization Error (LE)	Localization Recall (LR)	SELD Score
0.0	0.872	0.126	43.055	0.45	0.634
1.0	0.872	0.352	27.157	0.56	0.527
2.0	0.872	0.135	22.281	0.3	0.64
3.0	0.872	0.0	37.277	0.039	0.76
4.0	0.872	0.02	52.375	0.139	0.751
5.0	0.872	0.122	39.174	0.733	0.558
6.0	0.872	0.127	25.53	0.172	0.679
7.0	0.872	0.0	40.595	0.054	0.761
8.0	0.872	0.152	48.697	0.609	0.595
9.0	0.872	0.342	15.97	0.361	0.565
10.0	0.872	0.27	19.6	0.5	0.553
11.0	0.872	0.014	56.348	0.348	0.706
12.0	0.872	0.0	180.0	0.0	0.968

Figure 2: Class-wise Sound Event Localization and Detection (SELD) results on unseen test data of STARSS22 dataset. The table reports the Error Rate (ER), F-score (F), Localization Error (LE in degrees), Localization Recall (LR), and SELD Score for each sound event class.

The reproduced results on STARSS22 show both class-wise and aggregate (macro/micro) performance metrics on unseen test data. The SELD score, which combines both detection and localization performance, reaches 0.669 (macro) and 0.583 (micro), indicating moderate success under the reduced training setup. The Error Rate (ER) remains consistently high at 0.872, reflecting many missed or incorrectly identified events, while the F-scores are low, particularly in macro average (12.8%), suggesting limited detection precision and recall. In terms of localization, the Localization Error (LE) averages at 46.8°, and Localization Recall (LR) is 32.8% (macro), showing that while some sound directions are correctly estimated, many are still inaccurate.

2. Results on STARSS23

The reproduced results on STARSS23 similarly present both class-wise and overall (macro/micro) SELD performance. The SELD score reaches 0.674 (macro) and 0.578 (micro), reflecting consistent performance across classes under reduced training settings. The Error Rate (ER) remains high at 0.844, indicating frequent detection errors or omissions, while the F-score is relatively low, 11.5% (macro) and 21.2% (micro), suggesting that the model struggled with precision and recall in detecting overlapping sound events. Localization performance, captured through Localization Error (LE) and Localization Recall (LR), shows moderate accuracy, with LE averaging 44.7° (macro) and LR at 28.0%, improving to 33.9° LE and 50.7% LR on the micro level.

Overall, these results indicate a performance drop due to computational constraints (e.g., fewer epochs, smaller batch size), yet they affirm the model's capacity to handle realistic spatial sound scenes in STARSS22 and STARSS23 with a reasonable degree of success.

SELD Class-wise Evaluation Results (STARSS23)

Class ID	Error Rate (ER)	F-score (F)	Localization Error (LE)	Localization Recall (LR)	SELD Score
0.0	0.844	0.273	35.613	0.589	0.545
1.0	0.844	0.308	28.266	0.589	0.526
2.0	0.844	0.032	45.082	0.136	0.732
3.0	0.844	0.01	58.555	0.225	0.734
4.0	0.844	0.03	41.124	0.063	0.745
5.0	0.844	0.184	27.999	0.749	0.517
6.0	0.844	0.007	19.291	0.011	0.733
7.0	0.844	0.022	34.226	0.026	0.747
8.0	0.844	0.197	42.522	0.579	0.576
9.0	0.844	0.086	38.545	0.327	0.661
10.0	0.844	0.282	14.18	0.24	0.6
11.0	0.844	0.069	15.579	0.111	0.688
12.0	0.844	0.0	180.0	0.0	0.961

Figure 3 Class-wise Sound Event Localization and Detection (SELD) results on unseen test data of STARSS23 dataset. The table reports the Error Rate (ER), F-score (F), Localization Error (LE in degrees), Localization Recall (LR), and SELD Score for each sound event class.

REFERENCES

- [1] Mu, Da, Zhicheng Zhang, and Haobo Yue. "MFF-EINV2: Multi-scale Feature Fusion across Spectral-Spatial-Temporal Domains for Sound Event Localization and Detection." *arXiv preprint arXiv:2406.08771* (2024).
- [2] Politis, Archontis, et al. "STARSS22: A Dataset of Spatial Recordings of Real Sound Scenes with Moving Sources for SELD." DCASE Workshop (2022).
- [3] Politis, Archontis, et al. "STARSS23: Sony-Tau Realistic Spatial Soundscapes 2023 Dataset for SELD." DCASE Workshop (2023).

